

ROOT CAUSE ANALYSIS REPORT

Incident ID: RCA-PUMP-2025-001

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Revision: 1.0

Asset: P-101

Asset Type: Refinery Centrifugal Pump

Failure Mode: Bearing Failure

Incident Classification: Critical Reliability Event

Prepared By: Reliability Engineering Department

1. Executive Summary

On 14 March 2025, Pump P-101 experienced a severe increase in vibration, bearing temperature and motor current.

The Industrial AI Copilot generated a Critical Reliability Alert after detecting:

- Failure Probability > 90%
- Remaining Useful Life < 20 Days

Subsequent inspection confirmed advanced bearing degradation.

The bearing was replaced during a planned maintenance window, preventing an unplanned outage.

2. Asset Information

Asset ID:

P-101

Service:

Hydrocarbon Transfer

Duty:

Continuous Operation

Operating Hours Since Last Overhaul:

28,700 Hours

Criticality:

High

Production Impact:

High

3. Initial Alert

Date:

14 March 2025

Time:

09:42

Alert Type:

Critical Reliability Alert

Generated By:

Industrial AI Copilot

Severity:

Level 4

4. Telemetry Timeline

Day -30

Bearing Temperature:

67°C

Vibration:

3.1 mm/s

Motor Current:

108 A

Failure Probability:

28%

RUL:

187 Days

Condition:

Normal

Day -21

Bearing Temperature:

72°C

Vibration:

4.2 mm/s

Motor Current:

115 A

Failure Probability:

42%

RUL:

138 Days

Condition:

Early Degradation

Day -14

Bearing Temperature:

79°C

Vibration:

5.8 mm/s

Motor Current:

123 A

Failure Probability:

61%

RUL:

94 Days

Condition:

Progressive Wear

Day -10

Bearing Temperature:

84°C

Vibration:

6.9 mm/s

Motor Current:

131 A

Failure Probability:

78%

RUL:

57 Days

Condition:

Advanced Wear

Day -5

Bearing Temperature:

88°C

Vibration:

8.1 mm/s

Motor Current:

139 A

Failure Probability:

89%

RUL:

27 Days

Condition:

High Risk

Day -2

Bearing Temperature:

92°C

Vibration:

9.2 mm/s

Motor Current:

145 A

Failure Probability:

95%

RUL:

14 Days

Condition:

Critical

5. Operational Observations

Operators reported:

- Increasing mechanical noise
- Increased vibration near bearing housing
- Elevated motor load
- Reduced hydraulic efficiency

No seal leakage observed.

No cavitation symptoms observed.

6. Initial Diagnostic Assessment

The AI Copilot evaluated telemetry patterns.

Observed Pattern:

Bearing Temperature ↑

Vibration ↑

Motor Current ↑

Flow Rate Slightly ↓

Predicted Failure Mode:

Bearing Failure

Confidence:

96%

Predicted RUL:

14 Days

Recommended Action:

Immediate inspection

7. Investigation Activities

Reliability team performed:

- Vibration analysis
 - Thermal inspection
 - Lubrication inspection
 - Bearing housing inspection
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8. Vibration Analysis Results

Observed:

Dominant bearing defect frequencies.

Increasing vibration amplitude.

No significant shaft imbalance detected.

No coupling misalignment detected.

Conclusion:

Bearing defect confirmed.

9. Thermal Inspection Results

Observed:

Localized heating at drive-end bearing.

Maximum temperature:

94°C

Motor temperature remained within acceptable limits.

Conclusion:

Heat generation originated from bearing assembly.

10. Lubrication Assessment

Inspection Findings:

Lubricant degraded.

Presence of metallic particles.

Evidence of oxidation.

Reduced lubricant effectiveness.

Conclusion:

Lubrication deterioration accelerated bearing wear.

11. Bearing Inspection Findings

After disassembly:

Observed:

- Raceway pitting
- Surface spalling
- Heat discoloration
- Excessive wear

Bearing condition classified as:

Unserviceable

12. Root Cause Analysis

Method Used:

5 Why Analysis

Problem:

Bearing failure occurred.

Why 1

Why did the bearing fail?

Because excessive wear developed.

Why 2

Why did excessive wear develop?

Because friction increased.

Why 3

Why did friction increase?

Because lubricant effectiveness degraded.

Why 4

Why did lubricant effectiveness degrade?

Because replacement interval was exceeded.

Why 5

Why was replacement interval exceeded?

Maintenance planning did not account for accelerated degradation.

Root Cause

Primary Root Cause:

Lubrication degradation

Contributing Factors:

Extended maintenance interval

High operating duty

Elevated ambient temperature

13. Consequences

Potential consequences if left unresolved:

- Bearing seizure
- Shaft damage
- Seal damage
- Unplanned shutdown

Estimated avoided downtime:

18 Hours

Estimated avoided production loss:

Significant

14. Corrective Actions

Completed:

Bearing replacement

Lubricant replacement

Bearing housing cleaning

Condition monitoring calibration

15. Preventive Actions

Recommended:

Reduce lubrication interval.

Increase vibration monitoring frequency.

Implement lubricant condition monitoring.

Review PM strategy.

16. AI Copilot Performance Review

Failure Mode Prediction:

Bearing Failure

Actual Failure Mode:

Bearing Failure

Result:

Correct

Failure Probability:

95%

Assessment:

Accurate

RUL Prediction:

14 Days

Assessment:

Acceptable

Overall Assessment:

AI Copilot successfully identified failure progression before equipment failure occurred.

17. Lessons Learned

Key Lesson 1

Bearing temperature alone is insufficient.

The strongest indicator was the combined increase in:

- Bearing Temperature
 - Vibration
 - Motor Current
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Key Lesson 2

Lubrication degradation can accelerate rapidly under high duty conditions.

Key Lesson 3

Failure Probability and RUL together provide more actionable insight than either metric alone.

18. Knowledge Base Entry

Failure Signature:

Bearing Temperature ↑

Vibration ↑

Motor Current ↑

Root Cause:

Lubrication degradation

Corrective Action:

Bearing replacement

Severity:

Critical

AI Confidence:

19. Final Conclusion

The incident was caused by advanced bearing degradation driven primarily by lubricant deterioration.

The failure was successfully detected through telemetry analytics and AI-based predictive maintenance models.

Early intervention prevented an unplanned shutdown and avoided significant production losses.

This incident shall be retained within the refinery knowledge base and used by the Industrial AI Copilot for future diagnostic and recommendation activities.